



GCT
பொறிக்களம்

 **DCKAP**
INCUBATION CENTER

SIDE VIEW

FLIGHT CRAFT

RULE BOOK



Follow us On



porikkalam_gct



REGISTRATION FEES: Rs.199/-

TEAM SIZE: MAXIMUM OF 2 MEMBERS

PRICE POOL: Rs. 3250/-

Description

Design the Flight. Defy Gravity.

Take Engineering to the skies with **Flightcraft**, where imagination meets Aeromodelling. Design and build innovative flying machines, put your Aeromodelling skills to the test, and turn Engineering principles into flight. From the drawing board to the skies, every design is a test of Creativity, Precision and Performance.

Design. Build. Take Flight.





Round 1 – AERODYNAMICS QUIZ

Team Size

- Each team shall consist of 2 members.
- The same team members must participate throughout the event.
- No change of team members will be permitted between Round 1 and Round 2.

Quiz Format

- The quiz consists of **35** Questions carrying a total of **50** Marks.
- The Questions will be based on Aerodynamics.
- The Questions will be divided into the following sections:



Section	No of Questions	Marks per Question	Total Marks
MCQs	10	1 Marks	10 Marks
Fill in the Blanks – Formula Based	5	2 Marks	10 Marks
True/False & Visual/Image - Analytical Reasoning	5	2 Marks	10 Marks
Total	25 Questions	-----	30 Marks



Time Limit

- The total time allotted is 20 minutes.
- Teams must submit their Answers within the given Time limit.

Selection Criteria

- Based on the Scores obtained in Round 1, The Top 12 Teams will qualify for Round 2.
- The same Team members who participated in Round 1 must participate in Round 2.

Tie-Breaker

- In case of a Tie in scores, the team that submits the Quiz in the Shortest Time will be given preference.
- Submission Time will be considered as the Tie-breaker for selecting the Top 12 Teams.





Round 2 - GLIDER MAKING & FLIGHT

Participants must Design, Construct, and FLY a Glider using the materials provided by the Organizers, while adhering to the Specified Parameters. The following materials will be provided for each team to construct the Glider,

- Sun board
- Forks Sticks
- Flex Kwik
- Tape
- Cutter
- Rubber Band
- Clay





Parameters

- Maximum Wingspan: **500 mm**
- Maximum Overall Length: **400 mm**
- Maximum Allowable Weight: **100 g**
- Maximum Construction Time: **90 minutes**

Task

Participants must:

- Design and build a Functional Glider within the given Time limit.
- Use only the materials provided by the Organizers.
- Ensure that the Completed Glider complies with all Specified Dimensions and Weight limits.
- Successfully fly the glider during the Flight Evaluation.





Time Rule

Participants must complete and submit the Glider within the allotted 90 Minute Construction Time; No additional Construction or Modification will be permitted after the Time Limit.

Flight Trials

- Each Team will be given a Maximum of 3 Flight Trials.
- The participant must **personally launch** and fly their glider during each flight.
- The Best-performing Trial will be considered for the Final Score.
- No additional Trials will be permitted beyond the allotted 3 Attempts.
- Participants must follow the Instructions of the Event Coordinators during each Trial.
- Any unsafe constructions or Flying Practice may lead to Disqualification.





Evaluation Criteria

$$\text{Total Score} = R - 0.75D + 3T$$

- R (in meters) is the Range, Range is measured from the initial point to the point of first touchdown.
- D (in meters) is the deviation; Deviation will be measured as the perpendicular distance from the reference straight line.
- T (in seconds) is the time of flight. Time of flight will be measured from the time of launch to the time of first touchdown of the glider.

NOTE: The Highest Scorer will be the Winner.

FOR FURTHER DETAILS CONTACT:

DARSANI C S - 93630 25077

KIRUBANITHI B - 93453 54756